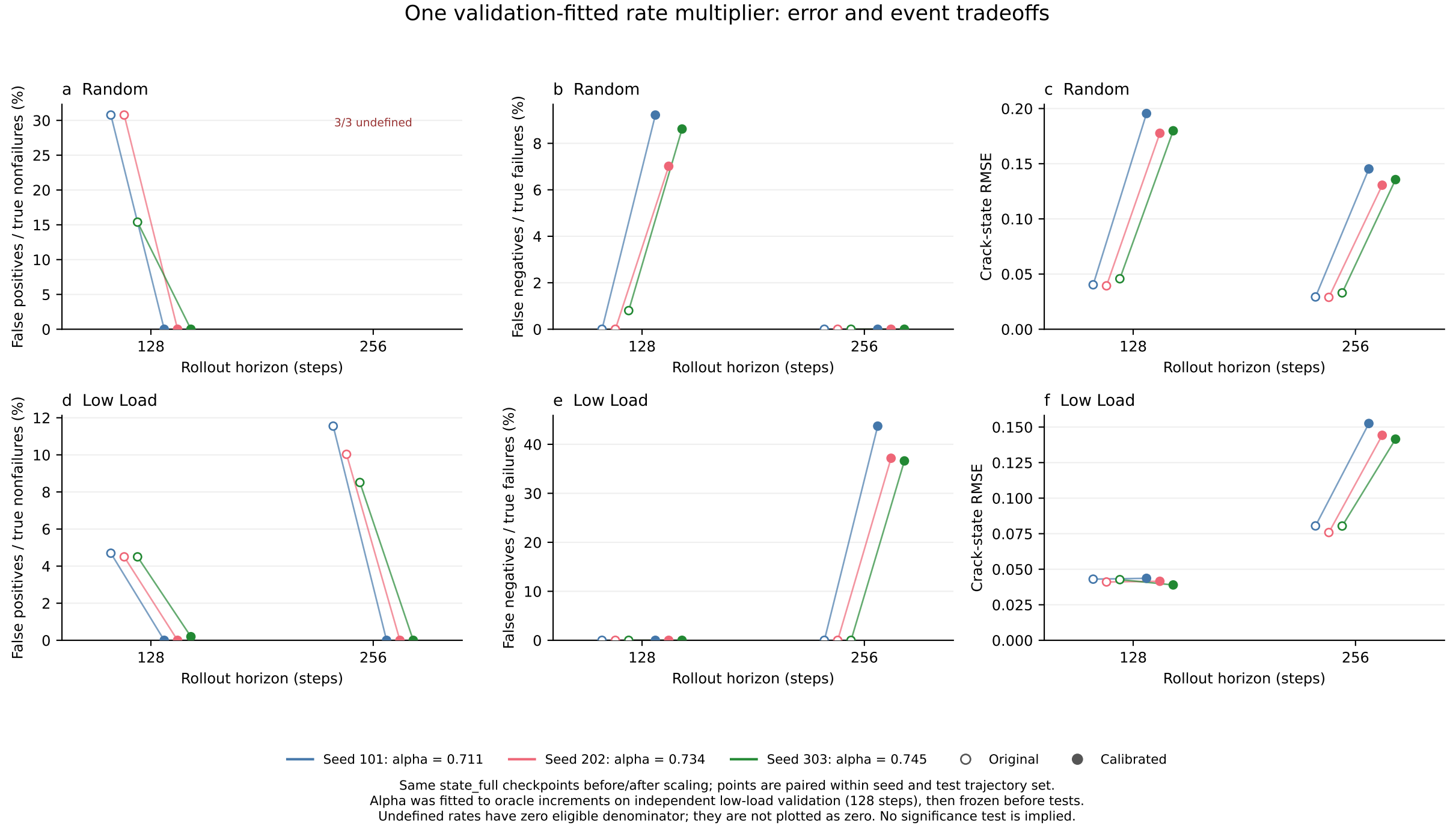
One validation-fitted rate multiplier: error and event tradeoffs
a Random
3/3 undefined
False positives / true nonfailures (%)
Rollout horizon (steps)
b Random
False negatives / true failures (%)
Rollout horizon (steps)
c Random
Crack-state RMSE
Rollout horizon (steps)
d Low Load
False positives / true nonfailures (%)
Rollout horizon (steps)
e Low Load
False negatives / true failures (%)
Rollout horizon (steps)
f Low Load
Crack-state RMSE
Rollout horizon (steps)
Seed 101: alpha = 0.711
Seed 202: alpha = 0.734
Seed 303: alpha = 0.745
Original
Calibrated
Same state_full checkpoints before/after scaling; points are paired within seed and test trajectory set.
Alpha was fitted to oracle increments on independent low-load validation (128 steps), then frozen before tests.
Undefined rates have zero eligible denominator; they are not plotted as zero. No significance test is implied.